

**1. Define scientific notation.**

A number written in scientific notation written as:

$$a \times 10^n, \text{ where } 1 \leq a < 10 \text{ and } n \in \mathbb{Z}$$

Here  $a$  is called the coefficient or base number.

**Note:**

- If the number is greater than 1, then  $n$  is positive.
- If the number is less than 1, then  $n$  is negative.

**2. What is the logarithm of a real number?**

The logarithm of a real number tells us how many times one number must be multiplied by itself to get another number.

The general form of a logarithm is:  $\log_b x = y$ . Where

- $b$  is the base,
- $x$  is the result or the number whose logarithm is being taken, and
- $y$  is the exponent or the logarithm of  $x$  to the base  $b$ .

This means that  $b^y = x$ .

**OR**

The logarithm of  $x$  to the base  $b$  is  $y$ , means that when  $b$  is raised to the power  $y$ , it equals  $x$ . The relationship between logarithmic form and exponential form is given below:

$$\log_b x = y \Leftrightarrow b^y = x \quad \text{where } b > 0, x > 0 \text{ and } b \neq 1$$

**3. Define common logarithm or Brigg's logarithm.**

If the base of logarithm is taken as 10 then logarithm is called common logarithm or Brigg's logarithm. It is written as  $\log_{10}$  or simply as  $\log$  (when no base is mentioned, it is usually assumed to be base 10).

**4. Define Natural logarithm.**

Logarithm having base  $e$  is called Napier logarithm or Natural logarithm.

$$\begin{aligned} \ln(0) &= \text{undefined} \\ \ln(1) &= 0 \\ \ln(e) &= 1 \end{aligned}$$

**5. Differentiate between characteristic and mantissa.**

The integral part of the logarithm of any number is called the **characteristic** and the decimal part of the logarithm of a number is called the **mantissa** and is always positive.

For example, if  $\log 278.23 = 2.4443$  then characteristic is 2 and mantissa is 0.4443

**6. Define antilog.**

The number whose logarithm is given is called antilogarithm. *i. e.* if  $\log y = x$ , then  $y$  is the antilogarithm of  $x$ , or  $y = \text{Anti } \log x$

In other words, **antilog is the inverse of a logarithm.**

**7. What is the difference between Common and Natural Logarithms?**

| Common Logarithm                                                                                                   | Natural Logarithm                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| The base of a common logarithm is <b>10</b> .                                                                      | The base of a natural logarithm is $e$ .                                                                                                   |
| It is written as <b><math>\log_{10}(x)</math></b> or simply <b><math>\log(x)</math></b> when no base is specified. | It is written as $\ln(x)$ .                                                                                                                |
| Common logarithms are widely used in everyday calculations, especially in scientific and engineering applications. | Natural logarithms are commonly used in higher-level mathematics, particularly calculus and applications involving growth/decay processes. |